Technical Service Bulletin

Technical Service Bulletin: TSB110084	**Released Date:** 04-Mar-2014
Water Pump Improvements	

Water Pump Improvements

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This document was originally released between 1994 and 2001. It has been added to Quickserve™ Online for informational purposes.

This document introduces product improvements to the coolant and oil seal of the water pump used on K38, K50, QSK45, and QSK60 engines. Recently, concerns have been raised about water pump seals leaking oil and water on these engines. Water seal leakage is generally due to filming deposits on the seal faces. This can be attributed to inadequate antifreeze concentration, poor coolant condition, poor water condition, increased levels of supplemental coolant additives (SCA), or the inability of the seal to disperse heat. Oil seal leakage is due to an excess of hot engine oil building up in the water pump drive and oil seal durability.

To address the water leak issue, a new seal has been developed for the water pumps. The sealing face spring load has been reduced by changing the spring rate and removing the reinforcement in the fabric from the seal. Both of these changes will reduce the frictional loads on the seal face and prevent overheating of the seal.

To address the oil leak issue, a change has been made to the water pump oil seal on K38, K50, QSK45, and QSK60 engines. The water pump oil seal is now a Viton™ seal, and this replaces the old polyacrylic material seal. This will improve the sealing capability and provide added durability.

The new water seal is 2 mm [0.0787 in] longer than the old water seal, and it is necessary to machine the existing water pump body to accept the new seal. The previous distance from the rear face of the water pump body (impeller end) to the water seal seat ledge was 134.87 mm ±0.25 mm [5.31 in ±0.01 in]. The new body dimension is 136.88 mm ±0.25 mm [5.39 in ±0.01 in].

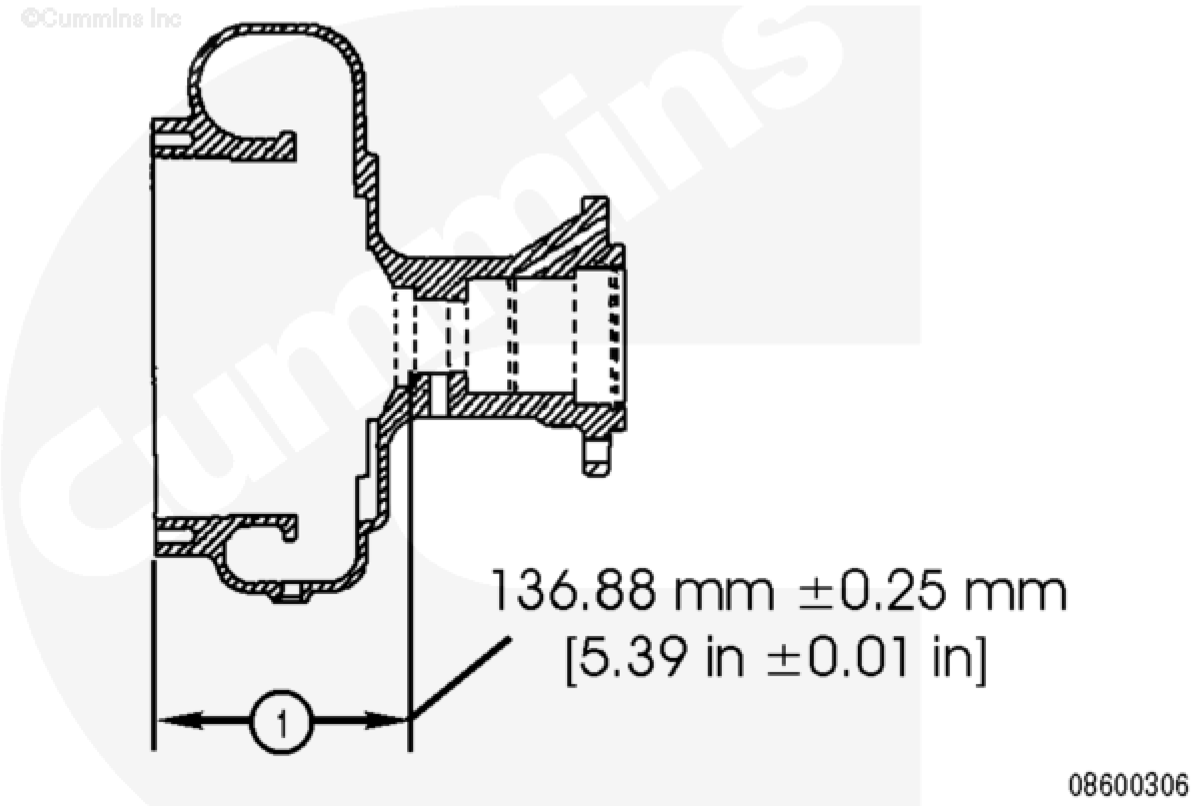

Figure 1, Machine Old Water Pumps to Accept New Seal Size

New Parts

- A new shaft, Part Number 3643961, has been released for water pumps on K38, K50, QSK45, and QSK60 engines. The new shaft is backward/forward compatible. The old shaft, Part Number 3626952 and Part Number 3634049, can continue to be used in new and existing pump bodies.
- The new water seal is similar in appearance to the old water seal. Reference Figures 2 and 3 below for a comparison view of the old and new water seal. Figure 2 shows the old water seal, Part Number 3089056, with the retaining ring having three notches holding the seal halves together. Figure 3 shows the new water seal, Part Number 3634007, with a one-piece retaining ring design.

⚠ CAUTION ⚠

Do not fit the new seal in an existing pump body. Doing so can cause rapid seal failure.

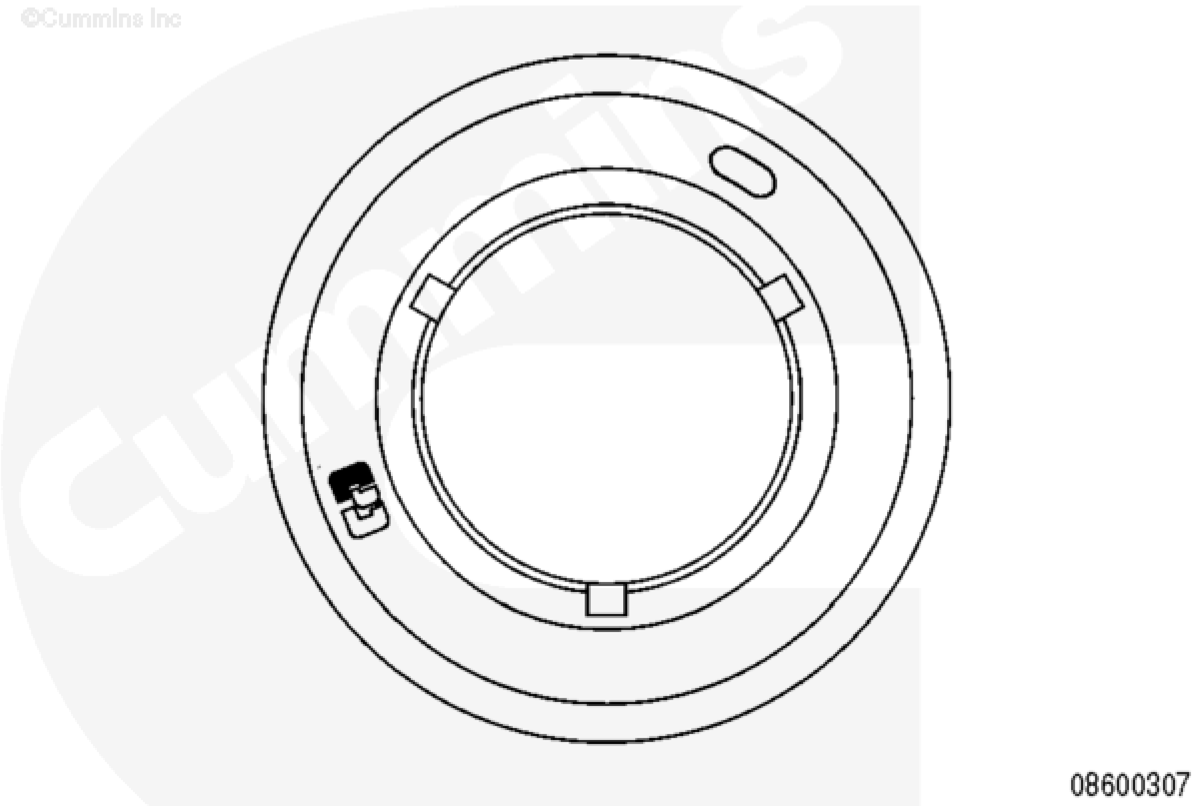

Figure 2, Existing Water Seal, Part Number 3089056

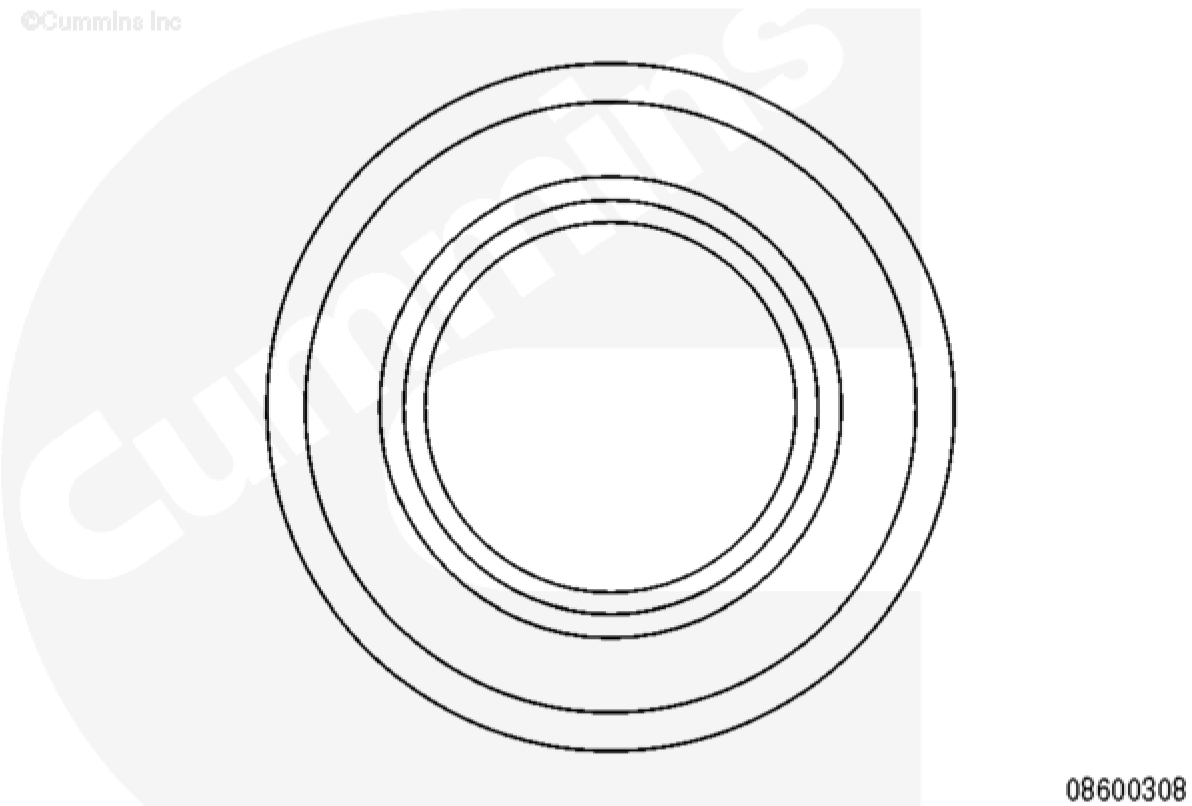

Figure 3, New Water Seal, Part Number 3634007

A new Viton™ oil seal has been released for K38, K50, QSK45, and QSK60 water pumps. The new seal is Part Number 3643960. The new seal is dimensionally identical to the previous seal, Part Number 3634146.

Rebuild

- The dimensions of a water pump body **must** be confirmed before rebuilding a water pump. Reference the information on page 1 of this document. If the present water pump is an old body as described above, it will be necessary to machine the body to the new specifications.
- Follow the rebuild instructions in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 008-062 in Section 8 (/qs3/pubsys2/xml/en/procedures/28/28-008-062-tr.html), with the following exception: Do **not** use water pump seal driver, Part Number 3824836. Use the new water pump seal driver, Part Number 3164175.

Table 1, Compatibility Matrix	
New Seal, Part Number 3634007	Old Seal, Part Number 3089056
New shaft, Part Number 3643961	New shaft, Part Number 3643961
Old shaft, Part Numbers 3626952 and 3634049	Old shaft, Part Numbers 3626952 and 3634049
New water pump body	Old water pump body
New seal driver tool, Part Number 3164175	Old seal driver tool, Part Number 3824836

Table 2, Engine and Water Pump Part Number Combinations

Engine	Description	Old Part Number	New Part Number	Remarks
K2000E, K1800E, K38	Water pump	3627083	3634029	Standard K38 water pump
K38	Water pump	3629785	3634040	K38 total flow is directed to the auxiliary heat exchanger
K38M0/1/2, K50M2	Water pump	3393398 3634037	N/A	Pumps with ports for emergency shutdown and with 29 degree oriented outlet flange
K50	Water pump	3627084	3634033	Standard K50 pump
K50	Water pump	3629787	3634043	K50 total flow is directed to auxiliary heat exchanger

Table 2, Engine and Water Pump Part Number Combinations				
Engine	**Description**	**Old Part Number**	**New Part Number**	**Remarks**
QSK45	Water pump	3410309	4065583	Internal or external with 34.474 kPa [5 psi] of restriction
QSK45	Water pump	3410313	4065589	Internal or external with 34.474 kPa [5 psi] of restriction
QSK45, QSK60	Water pump	3410314	4065594	Internal or external with 34.474 kPa [5 psi] of restriction
QSK60, QSK78	Water pump	3410311	4065586	Standard QSK60 and QSK78

Table 3, Kit Description		
Kit Part Number	**Type of Kit**	**Seal Part Number**
3803285	Major rebuild	3089056
3803284	Minor rebuild	3089056
4025005	Major rebuild	3634007

Document History

Date	Details
2011-3-22	Module Created
2013-5-5	Current version of the bulletin refers to the old obsoleted shop manual, bulletin 3810384.
2013-5-28	QSOL Quick Fix|Reason: Outfile|Notes: none
2014-3-4	none

Last Modified: 04-Mar-2014